

SHINING : README

Components

- 1 [Arduino Uno R3](#) or any board with the required pins (see the arduino code)
- 1 [Buck Boost ZK-4KX](#)
- 1 [12V 3A power supply](#)
- 1 [SSD-1306](#)
- 1 [KY-040 encoder](#)
- 2 [12V DC brushless mini pump](#) or any 12V DC mini pump that doesn't require to be submerged (a peristaltic pump would be best)
- 1 [3-24V buck module step down](#)
- 1 [28BYJ48 5V stepper + Driver ULN2003A](#)
- 2 [MOSFET irfz44n](#) (would also work with npn mosfets)
- 1 [Buzzer 3-24 V](#)
- 1 [galvanization tank](#)
- 1 breadBoard
- Dupont jumper cables (male-male/female-male)
- clear acrylic sheet (3-5 mm)
- [4x6mm clear polyurethane](#) tube
- 3 [10k Ohm resistors](#)
- 4 M4x16mm screws
- 2 M4x8mm screws
- 4 M4 Washers
- 4 M2 screws (for mounting the screen)

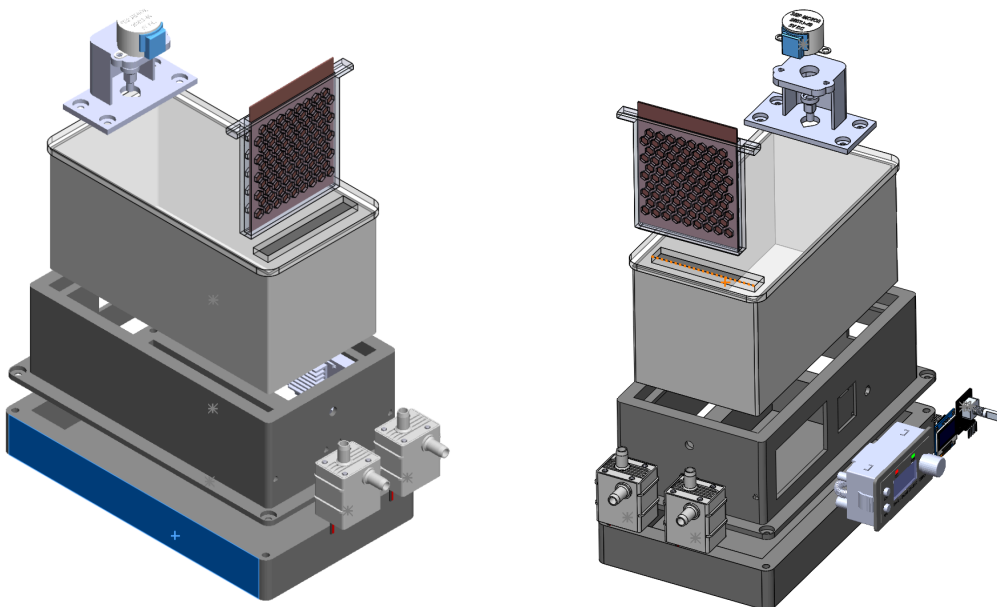
Consumables

- [Bright blue copper electrolyte](#)
- [pure copper anode](#) (or any pure copper sheet)
- Copper threads (1-2mm for holding the part and 0.3-0.8) for best conductivity)
- [conductive copper spray](#)

Equipment

- Standard 3D printer (PLA/TPU)
- Drill with 6mm and 12 mm bits (for the acrylic sheet)
- cutting tools for the acrylic sheet (dremel/handsaw/jigsaw...)
- hot glue gun

Assembly



As seen in those pictures the assembly is pretty straight forward :

- the main frame (box_shining) is screwed on the base with washers and the tank simply sit on top of the mainframe
- the stepper compartment (step_holder) is held by 4 screws going through the acrylic sheet in another 3D printed part (-step_holder_2) (not seen in the 3D view)
- the acrylic sheet can't be 3D printed, I added a DXF file that can be used as a guide when cutting or if possible, used for laser cutting the sheet
- the joint_TPU_Shining is glued on the acrylic and goes between it and the tank so that the lid does not move during use
- the tube is held in place at the bottom of the place by the embout_shining that is hot glued to the bottom wall (see image_embout)



For the electronic :

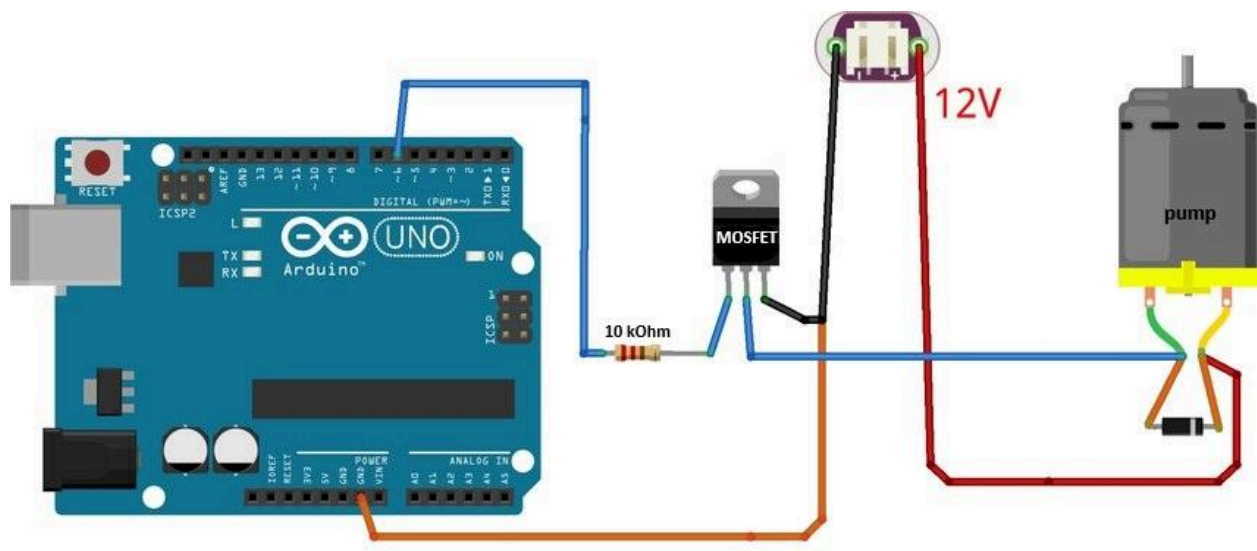
- the mini pumps are mounted with screws using an attachement, i'm using the one that came with mine but i will model a new one shortly
- the screen is mounted using m2 screws
- the encoder is already tapped and can be screwed directly on the frame
- the buck boost is a simple press fit
- Note that the power supply can't be properly put in the frame if the buck boost is all the way in. To mount it, get it in by the under side of the frame, press it against the inner wall of the frame and push the buck boost in. it should lock it in place (see image_up).

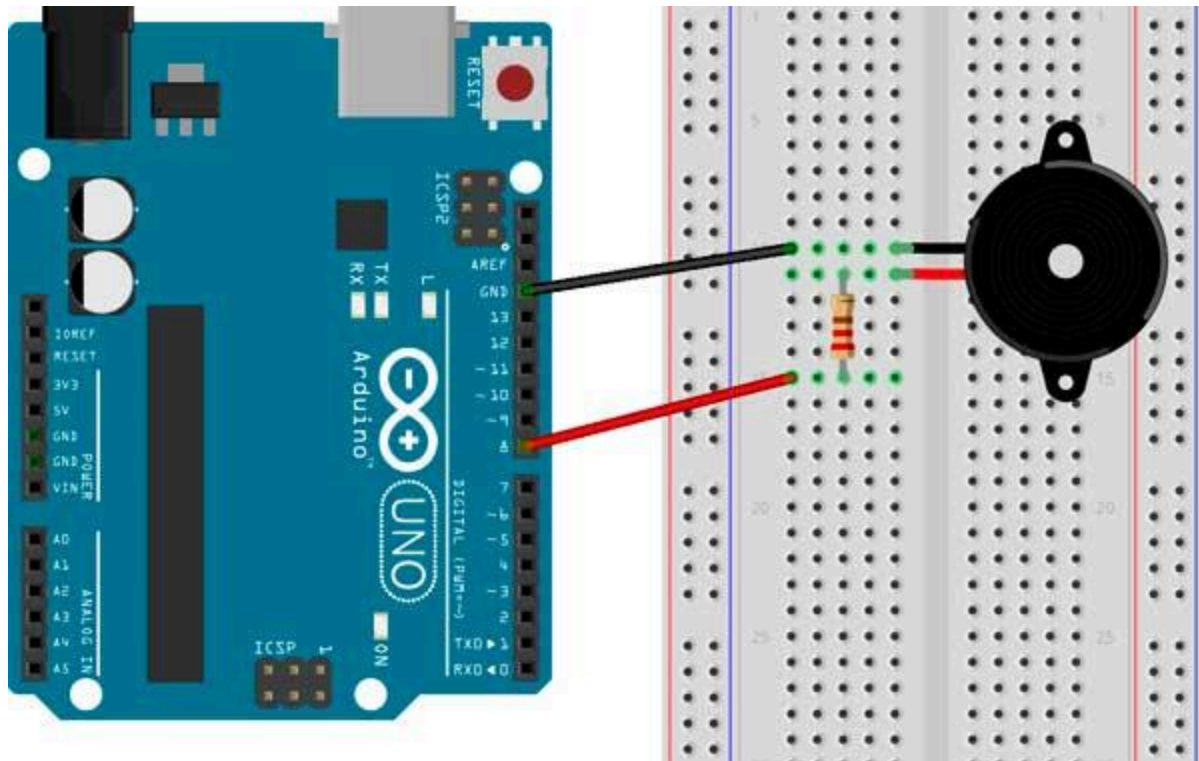


For the wiring I did not bother to make a tinkercad model since all the components are wired independently. Just follow the pictures for all the deferment components

Note that it uses 5-7V (power supply -> step down) to power the arduino, the step motor, the mosfets, the buzzer and the screen and 12V (direct from the power supply) to power the pumps and the buck boost. Do not mix those or you will damage your components.

Important : The pins may not be the same as in the picture, please refer to the arduino code





KY-040 Rotatory Encoder Connected with Arduino Uno

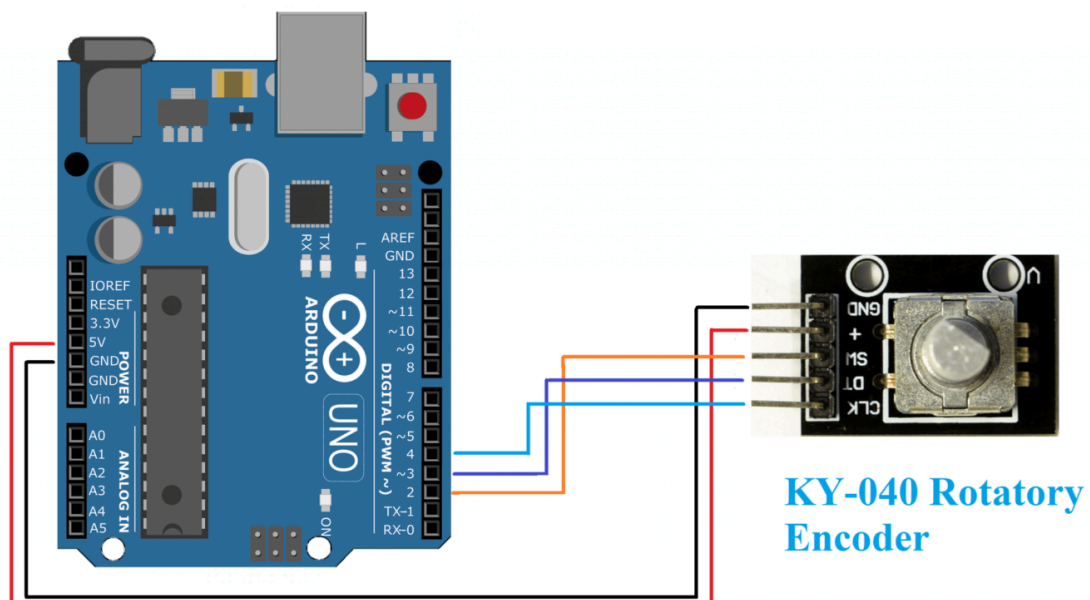
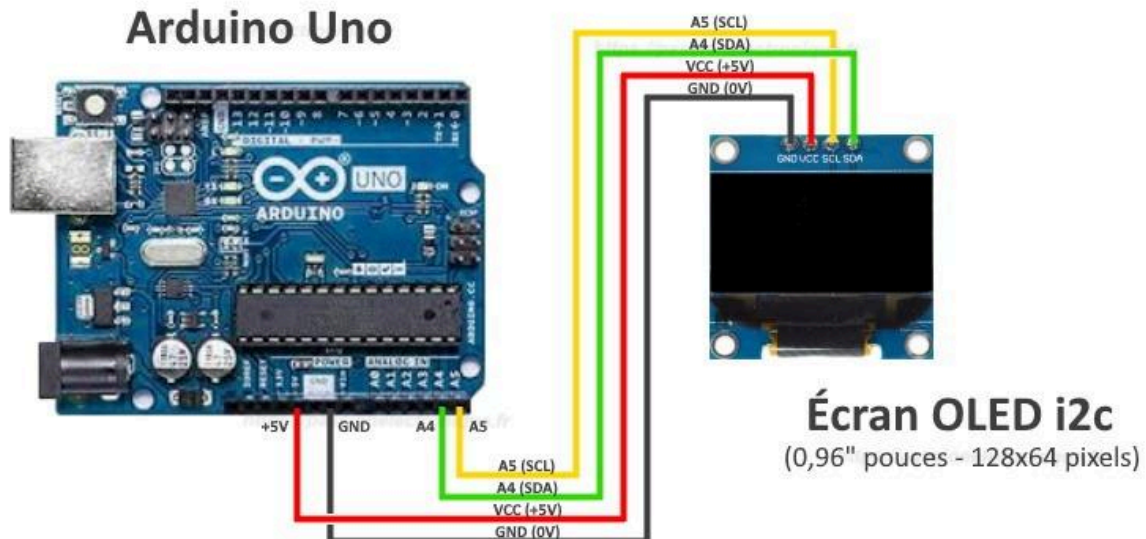


Schéma de câblage écran OLED sur Arduino Uno



PassionElectronique.fr

How to use

Preparation :

- sand the target part correctly to avoid any imperfection to show through the plating
- spray the target part with a conductive spray (see components)
- once dried, wrap your part with thin copper wire to ensure conductivity
- take the thicker wire to make a hook and attach it to the stepper
- attach the part to the hook inside the tank so that it does not exceed $\frac{3}{4}$ of its height
- connect the part to the VOUT - and the anode to the VOUT + check the connection with a multimeter
- check that the tubing is well connected between the tank, the pump and the bottle of acid

Use

- 1 : power the device by plugging it (there is not on/off button)
- 2 : use the encoder to choose a timer and click to confirm
- the device will first pump the electrolyte in the tank
- Use the buck boost to choose the current, if everything is well connected and the value is correct (1A per Dm^2) the constant current led should light up
- If you want to interrupt the process, click on the encoder
- Upon interruption or at the end of the timer, the second pump will get the acid back in its bottle
- using gloves, take the piece and rinse it with water to neutralise the last drop of acid.
- you can now sand the piece for better results